SIH 2026: Digital Twin Synthetic Data Protocol

Target Problem: SIH26120 (CSS & SRP Optimization)

Author: Bishes Sarkar | Roll: 23CH1001

1. Process Modeling & Thermodynamic Assumptions

This dataset fuses reservoir thermophysics with surface mechanical telemetry. The target variables are dynamically calculated based on foundational chemical engineering principles to provide a realistic simulation environment for machine learning optimization.

Viscosity-Temperature Dependency

Heavy oil viscosity decay during the soaking phase is modeled using Andrade's correlation for fluid dynamics. As heat energy disperses, production temperature T decreases, leading to an exponential spike in dynamic viscosity μ:

$$\mu(T) = A \times e^{B / T}$$

SRP Mechanical Constraints

Pump failure states (Rod Floating and Pump Unsetting) are classified synthetically using boundary constraints. Rod floating probability P_{float} approaches 1.0 when kinematic resistance exceeds the rod string's gravitational acceleration (high μ coupled with high Strokes Per Minute, Σ_{SPM}).

2. Data Dictionary (Feature Engineering for ML)

- **Well & Reservoir:** Depth (m), Permeability (mD), Initial Pressure (psi).
- **Fluid Properties:** API Gravity, Oil Viscosity (cP - dynamic target).
- **CSS Cycles:** Steam Temp (°C), Steam Quality (%), Injection Vol (bbl), Soak Days.
- **VFD / SRP Telemetry:** SPM (Strokes/Min), Stroke Length (in), Peak/Min Polished Rod Load (lbs).

- **Failure Flags (Classification Targets):** Failure_Rod_Float (0/1), Failure_Pump_Unset (0/1).
- **Production KPI (Regression Target):** Oil_Rate_bpd.

Note: The full dataset containing 5,000 continuous operating records has been exported as a separate CSV file for direct ingestion into Python/Pandas ML pipelines.

3. Dataset Sample Extract (First 40 Records)

Well_ID	Depth_m	Permeability_mD	API_Gravity	Init_Press_psi	CSS_Cycle	Steam_Temp_C	Steam_Quality_pct	Steam_Vol_bbl	Soak_Days	Prod_Temp_C	Oil_Viscosity_
BGH-138	1279.7	1417.8	15.6	1114.8	4	261.1	74.0	13010.2	5	174.1	226
BGH-128	1176.7	1403.7	15.1	1148.9	2	266.8	78.4	12578.2	11	165.4	258
BGH-114	1204.9	1287.1	14.8	1100.8	4	313.8	80.6	16795.4	13	193.8	170
BGH-142	1256.0	1770.4	14.9	1079.8	4	270.2	78.8	15209.9	12	163.2	267
BGH-107	1166.2	1342.4	15.5	1030.5	4	307.3	80.3	14045.4	8	198.4	160
BGH-120	1248.2	1598.1	16.0	989.8	5	270.0	73.8	16205.1	9	169.8	241
BGH-138	1096.9	1367.7	15.5	1098.4	1	315.8	82.5	19234.6	7	200.0	156
BGH-118	1227.5	1404.6	15.4	1130.8	5	314.0	76.7	15515.9	6	200.0	156
BGH-122	1278.6	1477.2	14.1	1058.4	2	253.0	84.8	15582.7	7	168.5	246
BGH-110	1171.0	1609.0	16.2	1152.8	5	303.9	70.3	14956.9	11	181.0	204
BGH-110	1255.9	1448.2	15.5	1065.6	5	261.9	75.7	14215.6	12	162.6	270
BGH-123	1291.7	1272.2	15.2	1154.2	3	265.1	72.7	16638.2	6	180.1	207
BGH-135	1187.3	1503.8	15.7	1061.0	2	282.2	78.3	14614.8	13	165.5	258
BGH-139	1183.1	1346.7	14.8	966.9	2	290.3	73.6	15999.8	8	190.6	178
BGH-123	1155.1	1808.8	15.2	1131.3	2	272.8	81.1	18001.2	6	182.6	199

Well_ID	Depth_m	Permeability_mD	API_Gravity	Init_Press_psi	CSS_Cycle	Steam_Temp_C	Steam_Quality_pct	Steam_Vol_bbl	Soak_Days	Prod_Temp_C	Oil_Viscosity_
BGH-102	1174.5	1448.0	15.6	1053.9	5	290.3	83.2	16489.5	11	178.5	212
BGH-121	1159.2	1366.4	15.0	1022.9	2	261.9	73.2	12452.6	6	176.2	219
BGH-101	1155.0	1895.0	14.2	1124.1	3	251.6	72.4	12335.6	11	154.6	307
BGH-123	1259.9	1647.8	14.1	1055.5	4	290.0	83.1	17506.6	12	179.5	209
BGH-143	1160.5	1320.6	15.9	1029.4	5	286.2	75.4	14775.6	13	161.3	275
BGH-129	1212.9	1341.8	15.5	1165.6	4	298.3	76.5	15521.7	5	190.0	179
BGH-137	1201.9	1594.8	15.4	1089.8	4	295.5	73.3	13416.3	6	196.2	165
BGH-101	1274.9	1508.7	14.6	1028.2	5	312.8	76.4	16232.1	6	196.5	164
BGH-120	1174.8	1678.0	15.5	1076.8	5	282.2	73.5	18775.7	11	170.8	238
BGH-132	1219.1	1367.5	15.2	1206.6	2	250.6	80.0	15303.2	10	150.8	327
BGH-111	1239.6	1640.2	15.0	1105.6	3	294.1	77.5	16295.7	13	177.0	216
BGH-121	1274.5	1052.3	15.4	1067.9	5	297.3	81.8	17490.9	7	189.3	181
BGH-143	1174.1	1469.1	15.7	1125.1	4	286.5	74.7	18360.3	12	173.9	227
BGH-124	1169.1	1376.6	16.9	1077.3	5	298.3	77.6	13641.5	9	176.2	219
BGH-148	1146.9	1409.0	16.0	1074.5	3	272.7	73.9	15827.3	9	167.6	249
BGH-126	1311.8	1479.4	16.2	1067.2	1	260.4	84.0	17550.3	9	173.7	227
BGH-141	1269.2	1727.7	16.4	1142.7	4	296.5	77.8	18108.2	12	186.9	187
BGH-127	1197.9	1269.2	14.9	1109.7	1	267.2	72.6	14152.6	11	168.7	245
BGH-115	1296.6	1062.4	15.2	1171.1	1	281.9	74.1	13659.7	11	180.5	206
BGH-114	1194.9	1292.4	14.8	1115.3	3	284.4	71.2	14690.0	8	177.8	214
BGH-146	1236.2	1704.5	15.0	1176.4	5	283.4	70.4	14358.5	10	180.3	206

Well_ID	Depth_m	Permeability_mD	API_Gravity	Init_Press_psi	CSS_Cycle	Steam_Temp_C	Steam_Quality_pct	Steam_Vol_bbl	Soak_Days	Prod_Temp_C	Oil_Viscosity_
BGH-143	1200.5	1171.0	15.2	1132.5	1	291.9	77.9	14442.7	5	193.5	171
BGH-102	1219.9	1385.6	14.9	1087.0	2	252.1	75.1	15823.9	10	151.3	324
BGH-136	1214.6	1667.9	15.7	1093.2	5	319.3	84.5	13918.4	9	191.5	176
BGH-106	1178.8	1488.3	15.8	1191.4	4	300.1	72.6	18118.3	5	185.9	190